# Advanced Lab Report:
# The train Line Noise Curve

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## Abstract

The telescope dome on top of Cudahy Science Hall is currently without a telescope, due to the vibrational noise from the Red Line trains which pass closely by, roughly every five minutes. In this work, we design and implement an experiment to characterize the excess noise signal generated by these trains, working towards the future implantation of a new telescope. We construct simple seismometers to detect the vibrational signal, which is then analyzed using power spectral density (PSD) estimates. We consider implications of our results and how our experimental design choices influenced these outcomes.

## I. MOTIVATION

The physics department of Loyola University Chicago (LUC), located in Cudahy Science Hall (CSH), has no working astronomical telescope installed in the dome which sits atop the building. From conversation with faculty in the department [4], we have learned that this is in part due to the vibrations from the nearby Chicago Transit Authority (CTA) train line. Specifically, this vibrational noise would constantly throw off the delicate calibration required for imaging of distant bodies unless an additional, expensive suspension system was installed to mitigate it.

This issue is not exclusive to LUC. For example, the LIGO observatory must account for noise from nearby train lines that affect the gravitational wave detector's data [5]. It is possible to similarly identify and account for vibrational noise produced by the CTA train line [1] that reaches CSH, with the goal to establish a working astronomical telescope in the building. Similar vibration studies have been performed before, leading us to believe that light-rail induced vibrations will still be a dominant signal even at distance from the train line [3]. Having access to hands on learning with such equipment would provide a deeper learning experience for the undergraduate students at LUC, and characterizing vibrational noise from the train Line is an important step in establishing an astronomical telescope at LUC. Our experiment also includes vibrational data the seismic monolith room, located in the bottom floor of CSH, to quantify any train noise the monolith detects and contribute to future local experiments on seismology.

To quantify vibrations CSH experiences due to nearby train activity, our group built three separate, GPS enabled, seismometers around the Arduino Uno R4 Wifi microcontroller in order to create frequency dependent noise plots from spatially separated probes. We also created power spectral density (PSD) plots of this data to analyze significant vibrations that would affect an astronomical telescope. We were able to implement a data collection an analysis pipeline to facilitate this process, although are currently unable to detect significant train-produced noise above background. It is important to note that this characterization would be only a first step and that we do not claim to currently be able to implement our findings into the installation of any telescope in CSH. However, results from these detectors could contribute to future prospects of establishing an astronomical telescope in CSH.

We structure this paper as follows: In section II we discuss the materials and equipment required to construct the experiment. In section III, we cover the design and construction of the seismometer network, including the amplifier circuit implemented. In section IV, we discuss the

| Component | Amount | Estimated Price |
|---|---|---|
| Arduino Uno R4 Wifi | 3 | $82.50 |
| Arduino GPS Shield | 3 | $89.85 |
| GPS Antenna | 3 | $64.50 |
| Antenna Adapter Cable | 3 | $11.85 |
| Micro SD Card (256 GB) | 3 | $239.97 |
| Portable Power Bank | 3 | $41.97 |
| Plastic Jar | 3 | $18.99 |
| 30 Gauge Magnet Wire | 3 | $38.94 |
| Neo. Magnet Set | 1 | $8.99 |
| Spring Set | 1 | $7.89 |
| Threaded Rod Set | 1 | $7.59 |
| Total (Retail) | N/A | $613.04 |
| Total (actual) | N/A | $530.54 |

FIG. 1: *(left)* Price summary for required components of the experiment. Note that prices vary based on manufacturer and retailer. *(right)* A photo of one seismometer on the LUC Physics Department's seismic monolith.

data collection, cleaning, and analysis methods used to produce our results, which are presented in section V. Finally, in section VI, we draw physical conclusions about our results and potential systematic error introduced by the design of our seismometer network.

## II. MATERIALS AND EQUIPMENT USED

To detect vibrations and convert them into workable data, we needed a number of materials. The Arduino components and other items we purchased to build our detector are listed in figure 1 below. In addition to these components, we also require the parts to make the amplifier circuit described in figure 2, which include the circuit components, wiring, breadboard, and soldering iron.

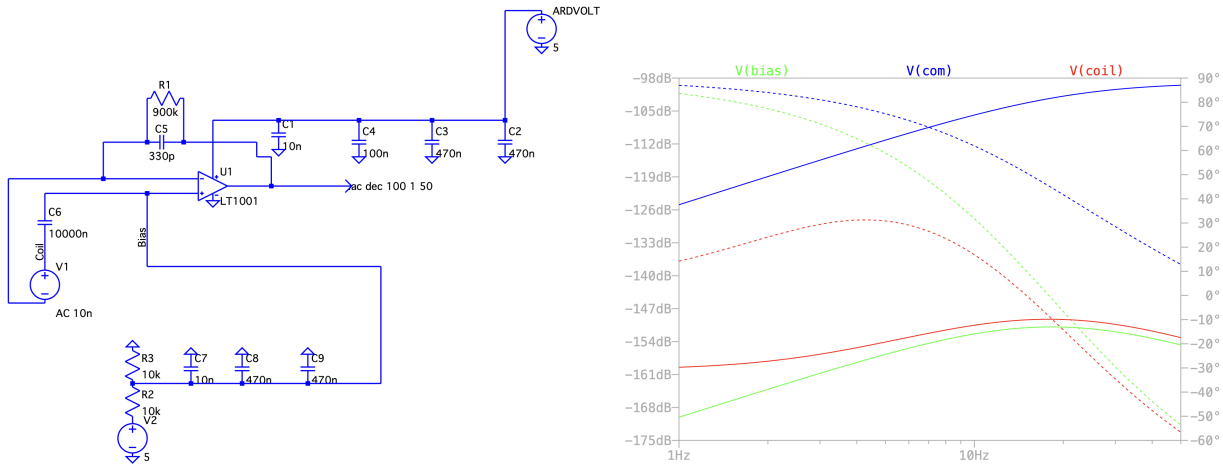


FIG. 2: Circuit diagram of amplifier circuit (left). Theoretical amplification of the circuit from LTspice (right). Blue represents amplified signal, red is the signal coming from the coil, and green is the bias voltage in order to avoid clipping in our final signal.

## III. DESIGN

Our group's first step was to design and construct three seismic sensors based around the Arduino Uno R4 Wifi microcontroller. To detect vibration, we built a geophone relying on a hanging magnet and a coil of copper wire. Inside of a plastic jar, we placed a coil of copper wire and suspended a magnet by a spring to hang in the center of the coil. We attached the coil to an amplifier circuit (figure 2) based around the LT1001 op amp, which is necessary to boost the small signal generated by the coil into something readable by our Arduino. We simulate the amplification produced by this circuit in LTspice for a small signal and present the resultant decibel plot in the right side of figure 2. From this circuit, the signal is then passed to the Arduino, which has an attached Adafruit Ultimate GPS Logger Shield to record accurate time and location.

The removable top of the jar allowed us to run the detectors without a magnet, which was helpful in characterizing the expected output. We observed a noise curve in the output, even when running with no magnet to generate the signal, which is representative of the inherent noise in our amplifier circuit (figure 3). Additionally, for detectors uno and tres, we notice that their amplifier noise curve shape matches the noise inherent in the LT1001 op amp [6].

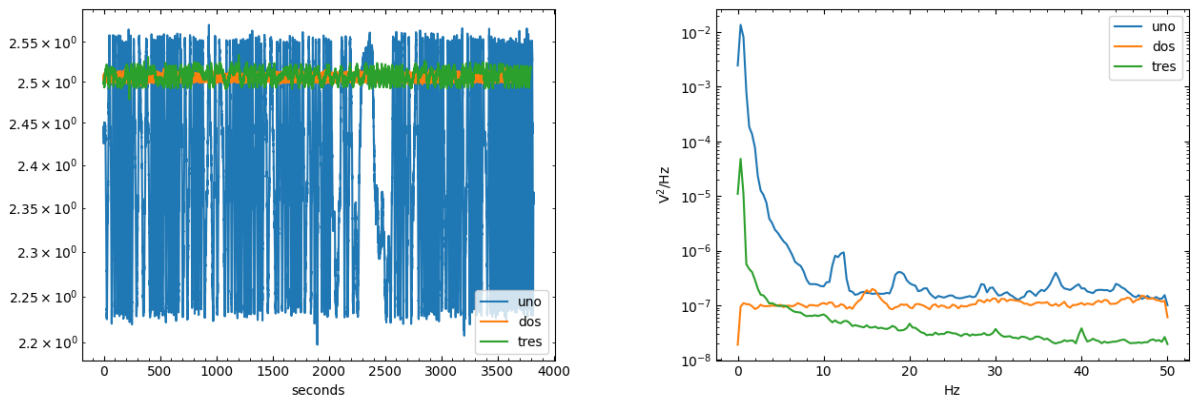


FIG. 3: The voltage of the no-magnet run is shown for all three detectors (left). A PSD plot of the same data shows the relative signal power with respect to frequency (right).

## IV. METHODS

Our goal was to have the detectors simultaneously record their location, time, and vibration sensor data to the SD card installed in the GPS shield. Along with representing the vibrations each detector experienced while trains were nearby, we recorded background noise outside of events for potential signatures during these times. These would all need to be analyzed above the inherent amplifier noise, which we discuss above.

With the magnets placed back into the detectors they were distributed to three separate locations: adjacent to the train line, near the dome of CSH, and on the seismic monolith at CSH. After starting the detectors, we manually recorded when trains were either stopping at or passing by the train station. Once the observing run had finished, we exported the data off of the SD cards and onto our Github repository for analysis.

We began our analysis in Python by cleaning and aligning the vibration data along the time axis. From here, we used our manually recorded data to identify events in the vibration data during which a train passing or stopping by. Windows were cut around these events so we could make a graph of vibrational data for each time a train was at the station, as well as graphs of when trains were not present to characterize background noise. We then used the Welch's method function included in the Python package SciPy [7] to compute an estimate of the PSD for each window, as well as for the data as a whole. We describe the utility of such a PSD and Welch's method for its estimation below.

A PSD plot, in plain terms, describes the strength of some signal in terms of its frequency

components. In signal analysis, these facilitate the recognition of resonant or dominant modes in a signal, which is especially useful for analysis of noise signals. One example of a PSD plot is the noise budget plots for the LIGO detectors in its fourth observing run [2], shown in figure **??**. Here, any signal at a given frequency is detectable if its strain is larger than the corresponding value on the noise plot. Displaying the detector noise in this frequency dependent way allows for easy, intuitive understandings of the areas where each detector will perform better or worse in.

In a similar way, we used a PSD plot to analyze the dominant parts of the seismic noise signal at the Lake Shore Campus (LSC). These will be calculated, as stated above, through the `scipy.signal.welch` function included in the SciPy Python library. This function takes a time series signal and breaks it up into overlapping segments, to which it applies a window function. Then, it computes the magnitude squared of the discrete Fourier transform of each segment, normalizing by the width of the segment. These results are then averaged to estimate the power density at each frequency, producing a PSD plot. For a more detailed explanation, see the original paper in reference [8].

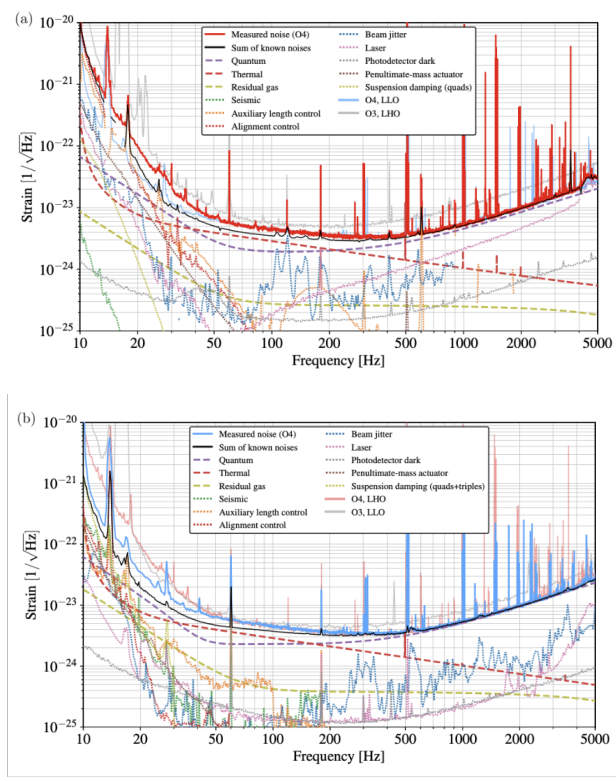


FIG. 4: The noise budget for the LIGO Livingston (lower) and LIGO Hanford (upper) detectors for the fourth observing run. Taken from reference [2].

The final signal analysis will need to consider the strength of the train event signal over the background noise. For this, we must convert from units of $V^2$/Hz to decibels (dB). As a refresher, for some characteristic power $P_0$ and signal power $P_0$, the signal strength in decibels is defined in equation 1. Using PSDs from detector runs with no magnets as our characteristic power allows us to determine the strength of any signal above this noise floor.

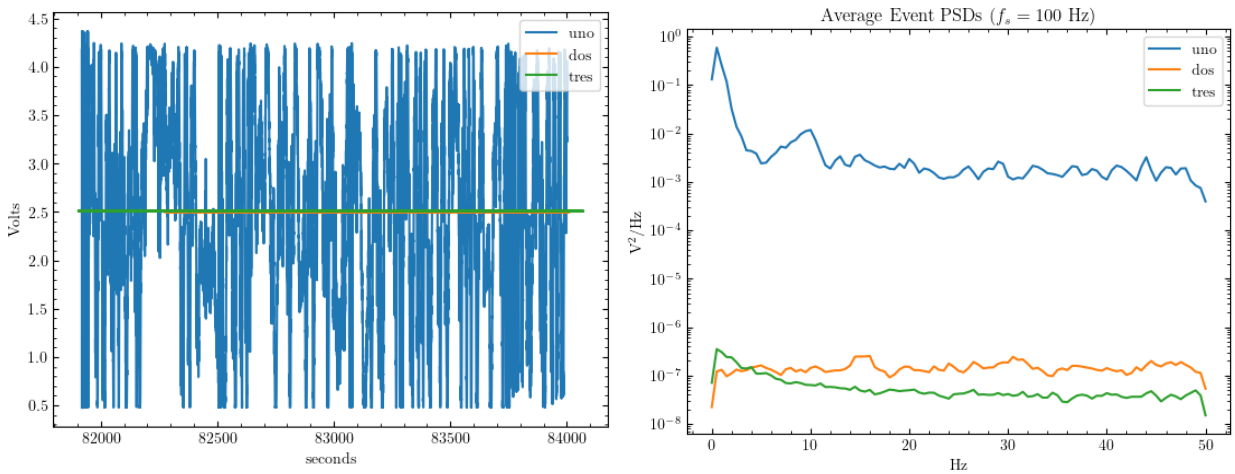


FIG. 5: *(left)* Raw voltage curves from our first concurrent data run. *(right)* Average of PSDs created from the manually selected event windows from our first concurrent data run.

$$dB = 10 \times \log\left(\frac{P}{P_0}\right) \tag{1}$$

## V. RESULTS

Our first concurrent run of the seismometer network was performed on a Sunday, when little foot traffic would interfere with our signal. The raw voltage curves and average of event window PSD estimates can be seen in figure 5.

Converting these PSD estimates to decibels and plotting in figure 6 reveals the true strength of the background noise signal over the amplifier noise and the strength of a train generated signal over this background.

The raw data and scripts used to produce these plots can be found in our project Github repository, available via request.

## VI. CONCLUSIONS

Currently, our seismometers are not able to detect the seismic noise signal from the train. This is most likely due to issues with the amplifier circuit, including impedance mismatches, issues with connections, or issues with clipping of the signal from too much amplification. Impedance issues can be solved with inductor based transformers or simply by adjusting the impedance of

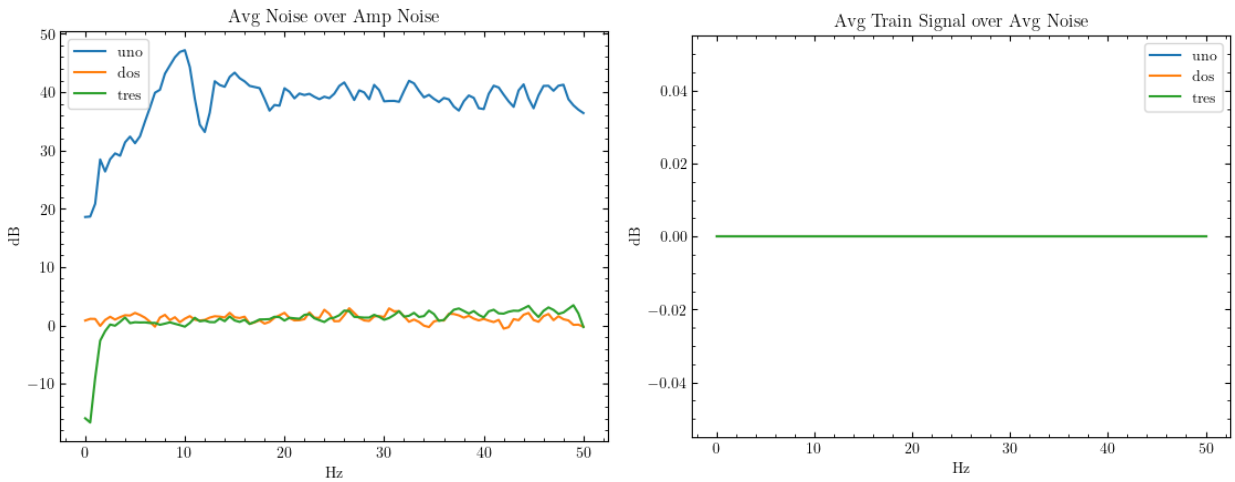


FIG. 6: *(left)* The PSD estimate for background noise as decibels over amp noise. *(right)* The PSD estimate for train signal as decibels over background noise.

both sides of the circuit. In order to rule out connection issues we will be going through the circuit with a voltmeter to diagnose the circuit. Finally, the clipping issue is simplest and only requires adjusting the resistor values in the amplifier circuit, so that the amplified output voltage is not above the op amp supply rails voltage.

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